Quantitative Comparison of Different Bi-Lexical Dependency Schemes for English

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Abstract

In this work, we summarize an in-depth quantitative and qualitative analysis of three dependency schemes for English, in order to acquire and document contrastive insights into how these schemes relate to each other. We demonstrate how a simple quantitative method can reveal salient structural properties of individual schemes, as well as uncover systematic correspondences between pairs of schemes. We use this method to estimate and compare the expressiveness of three dependency schemes and we identifiy several linguistic phenomena that have a different syntactic analysis in two of these schemes.

1 Introduction & Background

Bi-lexical syntactic dependencies have gained great popularity in the past decade, both as a comparatively theory- and language-neutral framework for syntactic annotation, and as the interface representation in syntactic parsing. For English (and maybe a handful of other languages), there exist different conventions for representing syntactic structure in terms of bi-lexical dependencies; variations on existing schemes as well as new ones emerge continuously. Despite great community interest in dependency syntax, there is relatively little documented knowledge about similarities and differences between the various annotation schemes (Ivanova et al. [4]; Zeman et al. [9]; Popel et al. [8]).

In this work, we summarize an in-depth quantitative and qualitative analysis of three dependency schemes for English, in order to acquire and document contrastive insights into how these schemes relate to each other (Eskelund [2]). We anticipate that such knowledge can (a) further our linguistic understanding of competing syntactic analyses; (b) support conversion across schemes and enable cross-framework evaluation; and (c) inform the design of future dependency schemes and treebanking initiatives.

We consider two common dependency schemes that are both derived from the syntactic analyses of the venerable Penn Treebank (Marcus et al. [6]), viz. Stanford Basic Dependencies (SB) (De Marneffe et al. [1]) and CoNLL Syntactic Dependencies (CD) (Johansson et al. [5]). As a more independent point of comparison, we include as a third annotation scheme what Ivanova et al. [4] term DELPH-IN Syntactic Derivation Trees (DT), based on a fresh annotation of the same PTB Wall Street Journal Text in the linguistic framework of Head-Driven Phrase Structure Grammar (Flickinger et al. [3];Pollard et al. [7]).

Our goals in this study are two-fold, viz. (a) to determine relative degrees of (dis-)similarity across different schemes and (b) to facilitate conversion between schemes, by uncovering systematic correspondences. For both objectives, we propose a quantitative, data-driven methodology, using simple descriptive statistics at various levels of detail. The data we have used for our study draws on Wall Street Journal Sections 00–17, which comprise 29,672 sentences (or 651,980 tokens) that can be aligned for tokens and PoS tags across the three schemes.

2 Comparison of Schemes and Scheme Pairs

To gain knowledge about the expressiveness of the dependency schemes, and possibly whether any of them are more expressive than the others, we will estimate the *granularity* and *variability* of each scheme. We define the granularity of a scheme as its available range of possible distinctions, which we contrast with variability, i.e. the amount of distinctions actually exercised. We presume that a scheme that is linguistically rich, will have a high degree of granularity and variability.

To estimate the granularity of the annotation schemes, we calculate the number of possible combinations of PoS tags and relation types for each scheme. According to available documentation, CD has the largest label set (69), DT the smallest (52), and SB falls in-between them (56). Assuming equal PoS tagging, CD thus has available the largest number of possible combinations of a relation label with a PoS assignment on either the head or dependent, or of course both. Conversely, we estimate the variability of the three schemes by counting labels and combinations of labels and PoS tags that are actually used in our data. The results, both in absolute and relative frequencies, are presented in Table 1.

	CD	DT	SB	CD	DT	SB
parts of speech	45	45	45			
dependency labels	62	50	49	89.9	96.2	87.5
head tag & label	546	588	677	17.6	25.1	26.9
dependent tag & label	688	690	577	22.2	29.5	22.9
dependent tag & head tag & label	3503	3541	3479	2.5	3.4	3.1

Table 1: Different tag and label combinations

We can see that only a small fraction of the possible combinations of PoS tags and labels are used. Although these proportions differ a bit between the different schemes, the numbers of combinations of dependent tags, head tags, and labels (shown in the last row) are surprisingly similar. This could indicate that the difference in variability between the schemes is trifling.

To further investigate variability across schemes, we consider the frequency distributions of the PoS tag and label combinations used, i.e. whether a small proportion of combinations covers a large number of dependencies, or whether these proportions are more evenly distributed. Table 2 shows how many tag–label combinations are required to 'cover' a given percentage of all dependencies in our treebanks. For all three schemes, the most frequently used combination of one PoS tag, for the head and dependent each, and one dependency type accounts for at least five percent of all dependencies; only a little more than 300 distinct combinations (or less than ten percent of all distinct combinations) cover at least ninety percent of the treebank. Again, these distributions are very similar across our three schemes, i.e. highly comparable in skewness and, thus, variability.

	5%	10%	25%	50%	75%	90%	95%	100%
CD	1	2	9	34	115	319	479	3503
SB	1	2	8	36	129	325	565	3479
DT	1	3	10	41	137	343	551	3541

Table 2: Tag and label combinations growth

We also investigate the tree-depth, the number of nodes in the longest path from the root to a terminal node in a sentence, in the three formats. The average (maximum) tree-depth in the SB scheme is 6.35 (20). Corresponding values are 7.75 (24) and 7.93 (25) for CD and DT, respectively. The SB scheme thus has a considerably lower tree-depth than the two other schemes.

Table 3 shows the 10 most common PoS tags for tokens used as roots and their percentage for each scheme. Table 3 confirms some of the facts we already know about the scheme. One of these facts is that in the SB scheme, content words are preferred as heads (De Marneffe et al. [1]). This explains that non-finite verbs forms (VB, VBG, VBN), nouns (NN) and adjectives (JJ) occur as roots far more often in the SB scheme than in the other schemes, while modal verbs (MD), in contrast to the other schemes, hardly ever are used as roots in SB. It can also explain why, as we can see from the table, there is a higher variation among PoS tags frequently used as roots in SB than in the other schemes.

	CD	DT	SB
VBD	43.5	38.3	35.8
VBZ	28.3	24.2	15.7
VBP	14.3	11.6	7.4
MD	8.2	7.1	0.1
CC	0.0	13.3	0.0
VBN	0.5	0.5	13.3
VB	0.7	0.6	8.2
NN	1.1	0.8	5.6
JJ	0.1	0.1	5.0
VBG	0.1	0.0	4.0
	96.8	96.5	95.1

Table 3: Most common tags of roots

Another phenomenon that Table 3 reveals, is that DT treats coordination different from SB and CD (Ivanova et al. [4]). Coordination is a well-known area of differences between dependency schemes. According to Popel et al. [8], three main models for this problematic issue are most frequently used: the Stanford parser style, the Mel'čuk style, and the Prague Dependency style. The SB scheme uses the Stanford parser style and CD makes use of the Mel'čuk style. The model used by the DT scheme is the Prague Dependency style, the only model among these three where the coordinating conjunction is considered the head of the coordination structure. From Table 3 we see that the coordinating conjunction (CC) often appears as root in DT, in the other schemes it practically never does.

We have also compared pairs of schemes, to gain knowledge about similarities and differences between them. Table 4 shows the unlabelled attachment score (*UAS*) and the unlabelled sentence accuracy (*USA*), both including and excluding punctuation tokens, as well as the share of identical roots for our scheme pairs. There is a considerably higher correspondence between roots in the DT/CD pair than in the other pairs. The UAS is quite similar for the DT/CD and the CD/SB pairs when punctuations are included, and substantially lower for the DT/SB pair.

Leaving out punctuation tokens gives a considerable increase in the similarity scores for both the SB/DT and the DT/CD pair. The scores for the CD/SB pair is nearly unchanged, showing that punctuation tokens are, in most cases, identically attached in these two schemes. The most similar schemes, when we exclude punctuation tokens, are DT and CD, even though—unlike CD and SB—DT does not derive from the original phrase structure annotations in the PTB.

	identical roots	w/ punctuation		w/o punctuation	
		UAS	USA	UAS	USA
CD/SB	62.5	72.7	13.1	72.9	13.1
SB/DT	53.7	55.7	1.2	60.5	5.3
DT/CD	84.4	73.6	1.2	80.8	17.7

Table 4: Similarity scores for scheme pairs

3 Detecting Structural Differences between Schemes

We know that the three schemes use different approaches to coordination structures, that content words are preferred as heads in SB, that CD uses non-projective dependencies to handle discontinuous structures and that punctuations are attached differently in DT than in CD and SB. Are there other systematic differences of syntactic structure, and how can we identify them? We have explored the use of a quantitative methodology for detection of patterns of different syntactic analyses in scheme pairs.

dependent tag	DT label	DT head tag	aligned	unaligned
CC			1029	14457
		CC	12125	20373
POS			43	5868
		POS	12	5997
	NUM-N		32	6682
CD			11317	12818
		CD	2010	15143
		DT	707	2398

Table 5: Distributions of aligned vs. unaligned dependencies (without punctuation)

We count aligned and unaligned dependencies per dependent PoS tag, *scheme*$_1$ label, *scheme*$_2$ label, *scheme*$_1$ head PoS tag and, for unaligned dependencies, also *scheme*$_2$ head PoS tag. We compare these data to see if we can find combinations of PoS tags and labels that differ in a way that can imply a systematic difference between the schemes. Subsequently, we extract sample sentences that adhere to these combinations and see if we can discover patterns of systematic differences. Finally, we rewrite the structures in *scheme*$_1$ to the *scheme*$_2$ pattern and recalculate the unlabelled attachment score.

We explore the use of this methodology, by applying it to the DT (*scheme*$_1$) and

CD (*scheme*$_2$) pair. We will concentrate on investigations of structures that constitute the unaligned dependencies with the dependent PoS tags, source format labels and/or head PoS tags shown in Table 5. The large number of unaligned dependencies involving CC (coordinating conjunction) heads and/or CC dependents, are caused by the use of different coordination models in the two schemes. Rewriting these structures in the DT data so that they adhere to the Mel'čuk style, increases the UAS for the DT/CD pair by 4.5.

A closer investigation of dependencies involving tokens tagged POS (*possessive ending*) reveals the information shown in Table 6. We see that most POS tokens are differently attached in DT and CD and the greater part of these are attached by a dependency relation labelled SP-HD in DT. The counts also reveal that most dependents attached to a POS head in DT, have a different head in CD. The greater part of these dependents, as well, are attached by arcs labelled SP-HD in DT.

dependent tag	DT label	DT head tag	aligned	unaligned
POS			43	5868
POS	SP-HD		13	5516
		POS	12	6031
	SP-HD	POS	12	5621

Table 6: Aligned vs. unaligned possessives

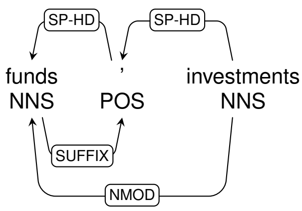

Figure 1: Example dependencies involving possessive endings (DT top, CD bottom)

Examining sample sentences, we find that the pattern illustrated in Figure 1 is common. While CD attaches the possessive ending to its noun (the possessor), DT treats it more like a two-place relation. Rewriting these structures to adhere to the CD pattern, increases the UAS for the DT/CD pair by 1.7 (punctuations are included in this score). We investigate the other groups of unaligned dependencies in the same manner. A closer look at NUM-N labels reveals that most of the dependents attached by an edge labelled NUM-N in DT, are attached to a different head in CD. We also find that most of these dependents are attached to a CD node in DT and that the greater part of them are tagged NN or NNS.

We have a closer look at some of these sentences and find the common pattern exemplified in Figure 2. DT seems to assign the amount to be head of the measure unit, while CD takes the opposite stance and chooses the measure unit as head of the amount. Rewriting these structures in the DT data, increases the UAS of the scheme pair by 2.0.

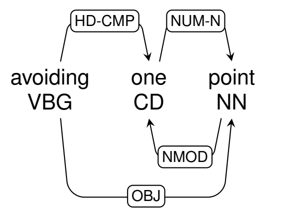

Figure 2: Example dependencies with NUM-N in DT

Examining unaligned dependencies involving CD (cardinal number) tokens further, shows us that there is a considerable number CD tokens that are attached to another CD token in our DT data, but attached to a '$' token (dollar sign) in the CD scheme. All these tokens are attached by an arc labelled SP-HD in DT. These structures, i.e. '$ 212 million', contain both a cardinal number and a numeral, both tagged CD. While DT attaches the numeral to the dollar sign and the cardinal number to the numeral, CD assigns the dollar sign as head of both the numeral and the cardinal number. Rewriting these structures increases the UAS by 0.5.

Further investigation of the remaining unaligned dependencies involving heads tagged CD in DT, reveals another pattern of systematic differences between the two schemes. In constuctions like 'until Dec. 31', DT assigns the day number as the head of the month. In CD the day number and month are not directly connected. Rewriting these structures increases the UAS by 0.4.

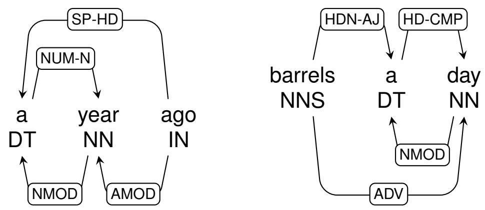

Figure 3: Examples of dependencies with noun dependents and determiner head in the DT data

We have also investigated the comparatively modest number of unaligned constructions that have determiner heads in DT. We find that more than half of these dependents are nouns. A closer look at them reveals several patterns, where DT assigns the DT node as head of the noun, while CD treats it as an ordinary determiner and considers it a dependent of the noun. Two patterns are shown in Figure 3. In the first of these examples, *'a'* is actually a cardinal number (meaning *'one'*), not a determiner. In the second graph *'a'* works as a preposition and should ideally have been tagged IN. Rewriting these structures increases the UAS for the format pair by 0.4.

4 Conclusions

We have demonstrated how a simple quantitative method can reveal salient structural properties of individual schemes, as well as uncover systematic correspondences between pairs of schemes. We have used this method to estimate and compare the expressiveness of three dependency schemes. We have also identified and documented several linguistic phenomena that have a different syntactic analysis in CD and DT. These and similar findings, together with our discovery procedure based on simple contrastive statistics, was applied by Eskelund [2] in the development of a heuristic converter from the DT to the CD scheme.

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